

Analysis and Mitigation of the AYE Westbound Weaving Bottleneck

CE5203 Traffic Flow and Control — Course Project, Spring 2026

Group 12

Leong Sio Kuan (A0328792N) Luo Wenjie (A0331290N)

Wang Xu (A0339365U) Zhao Jiaqing (A0331529E)

Department of Civil and Environmental Engineering, National University of Singapore

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Abstract

We study a recurring bottleneck on Singapore’s Ayer Rajah Expressway (AYE) westbound between Exit 9 (Clementi-bound on-ramp) and Exit 11 (Clementi off-ramp). A 46-min peak-period and 62-min off-peak field survey was conducted; vehicles were counted by class with a YOLOv11 pipeline at four measurement points. The empirical capacity of the 373 m weaving section is approximately **5,060 veh/h**, with peak-period total upstream flow of 5,060 veh/h (51.2 % car exit ratio, 1,785 veh/h ramp-merge flow) and off-peak flow of 5,050 veh/h (40.6 % exit ratio, 1,185 veh/h ramp-merge). A SUMO microsimulation calibrated to the YOLO speed/demand data was used to compute the four traffic performance parameters required by the project brief and to evaluate three control strategies — Variable Speed Limit (VSL), Ramp Metering (RM), and a combined VSL+RM — across both peak and off-peak demand profiles, plus a literature-recommended upstream VSL placement, giving **12 simulation runs**. Ramp metering at a 20 % downstream-occupancy threshold is the only consistently beneficial strategy: peak-period network total time loss decreased by **22.7 %** and average delay on the weaving section by **7.7 %**, with negligible effect during off-peak. In-section VSL on a single weaving lane degraded performance in both periods; relocating VSL upstream to the mainline removed most of the harm but did not match RM. We recommend deploying ramp metering alone on this site.

Keywords: weaving bottleneck, ramp metering, variable speed limit, microsimulation, SUMO, ALINEA, AYE Singapore.

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1 Introduction

The Ayer Rajah Expressway (AYE) connects Singapore’s central business district to the western residential and industrial districts [1, 2]. During evening peaks, severe congestion is observed on the westbound carriageway between Exit 9 and Exit 11, where a 1 lane on-ramp from Clementi/Buona Vista feeds traffic into the mainline immediately upstream of the Exit 11 (Clementi) off-ramp. The two opposing lane-change movements — merging vehicles from lane 0 and exiting vehicles from lanes 2 and 3 of the upstream mainline — compete within a 373 m weaving section, producing speed drops, queue spillback and elevated travel times.

Project task. Following the CE5203 brief, this project: (a) identifies the source of congestion; (b) computes the capacity of the system during the peak period; (c) computes at least two of the four required parameters (we report all four: average delay, average queue length, average time in queue, average excess accumulation); (d) repeats (b) and (c) for the off-peak period; (e) proposes and evaluates a control strategy in both periods.

Approach. The bottleneck location was first identified from LTA OneMotoring traffic-condition data and a field survey from College Link Road. A digital camera at a footbridge and a smart-phone facing the exit recorded 46 minutes of peak traffic on 19 March 2026 and 62 minutes of off-peak traffic on 24 March 2026. A YOLOv11 vehicle-counting pipeline extracted per-minute flows by class. A SUMO microsimulation of the weaving section, calibrated to the YOLO data, was then used to compute the four performance parameters and to evaluate three control strategies (VSL, ramp metering, VSL+RM) plus an upstream-VSL alternative.

Figure 1 shows the network layout reproduced in SUMO. The weaving section between the merge node and the diverge node contains four physical lanes (the additional left-most lane originates from the on-ramp); the upstream mainline (E_main_in) and downstream mainline (E_main_out) carry three lanes each.

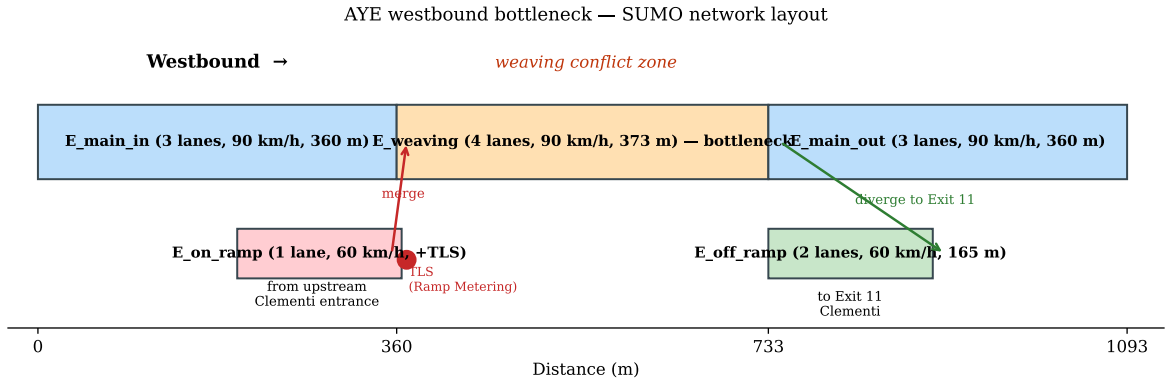


Figure 1: AYE westbound bottleneck reproduced as a SUMO network. The 373 m weaving section between merge and diverge nodes is the primary bottleneck. Length, lane count and speed limit per edge are shown.

2 Field Survey and YOLO Vehicle Counting

2.1 Field Survey

The peak survey ran from 18:31–19:14 on 19 March 2026 (3 minutes of low-quality data discarded, 43 valid minutes); the off-peak survey ran from 14:05–15:07 on 24 March 2026 (2 minutes discarded, 60 valid minutes). The mainline speed limit is 90 km/h (60 km/h for buses/coaches/large trucks); ramp speed limit is 50 km/h. Heavy vehicle composition was small in both periods.

2.2 YOLO Vehicle Counting

We use the YOLOv11 pre-trained model (chosen over the latest YOLOv26 for compatibility with the laptop GPU) to detect cars, motorcycles, buses and trucks frame by frame. Virtual detection lines on the video acted as loop detectors. To reduce double-counting we grouped overlapping detection lines so that a vehicle could only register once per group. Counts at the merge and exit measurement points were corrected using the formulas

$$\text{Exit volume} = V_{\text{exit lane (line)}} + V_{\text{LC-left after line}}, \quad (1)$$

$$\text{Mainline after exit} = V_{\text{mainline (line)}} - V_{\text{LC-left after line}}, \quad (2)$$

where $V_{\text{LC-left after line}}$ is the count of vehicles that performed a leftward lane change downstream of the detection line. The processed counts are summarised in table 1; the per-minute demand profile generated from the YOLO data is shown in figure 2.

Table 1: Field survey summary (excluding motorcycles).

Parameter	Peak	Off-Peak	Δ	Unit
Survey duration	43	60	–	min
Total upstream flow	3,627	5,050	–	veh
Flow rate	5,060	5,050	–0.2%	veh/h
Mainline flow	3,276	3,865	+18%	veh/h
On-ramp merge flow	1,785	1,185	–34%	veh/h
Exit flow	2,393	1,774	–26%	veh/h
Exit ratio (cars)	51.2 %	40.6 %	–	–

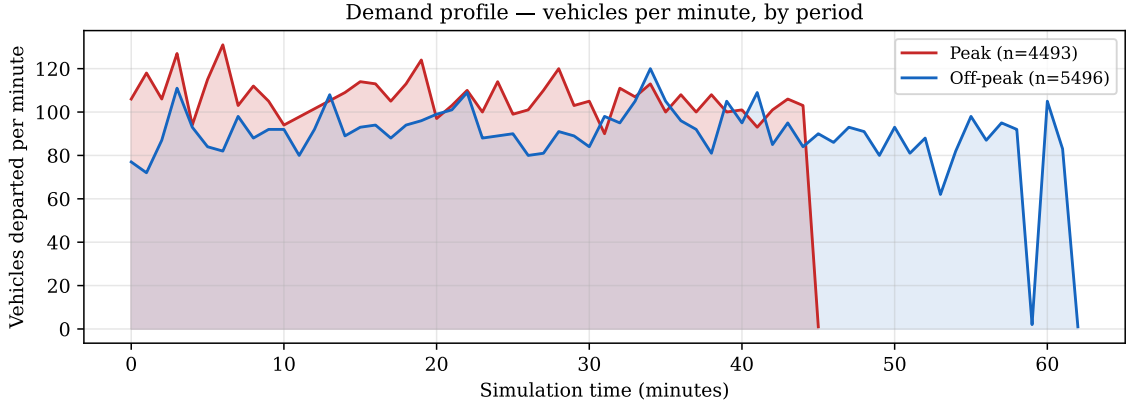


Figure 2: Demand profile per minute, derived from YOLO counts. Peak demand spans 0–45 min with a maximum of about 110 veh/min; off-peak demand spans 0–62 min with a more uniform profile.

3 Analysis During the Peak Period

3.1 Source of Congestion

The primary bottleneck is the 373 m weaving section between the on-ramp merge node and the off-ramp diverge node. Three conflicting movements coexist on it:

1. merging vehicles entering at lane 0 must change lanes rightward to join through-traffic;
2. exit-bound vehicles need to change lanes leftward to reach lane 0 or lane 1;
3. through-traffic competes for the remaining lanes.

During the peak survey, 51.2% of cars exited and the upstream on-ramp merged a flow of 1,785 veh/h — a 50% increase over off-peak. The combined lane-change demand exceeds the section’s ability to accommodate smooth weaving and queue spillback propagates upstream.

3.2 Capacity Estimation

Capacity was estimated from the YOLO upstream count over the peak survey:

$$\hat{C}_{\text{peak}} = \frac{N_{\text{upstream}}}{T_{\text{survey}}} = \frac{3,627 \text{ veh}}{43/60 \text{ h}} \approx 5,060 \text{ veh/h.} \quad (3)$$

The downstream measurement gave 5,231 veh/h, which agrees with equation (3) to within typical YOLO counting error. Off-peak total upstream flow was 5,050 veh/h over 60 min, essentially identical, indicating that the section operates near its empirical capacity in both periods (table 1).

3.3 SUMO Microsimulation Setup

A SUMO model of the network in figure 1 was built with seven nodes, six edges and nine lane-level connections (tables 2 and 3). The SUMO Krauss car-following model was used (default in SUMO), and the SL2015 sublane lane-changing model was activated by setting `lateral-resolution = 0.8m`, dividing each 3.2 m lane into four lateral strips.

Table 2: Edge topology.

Edge ID	Lanes	Length (m)	Speed (m/s)	[km/h]	Function
E_main_in	3	360	25.00	[90]	Upstream mainline
E_on_ramp_1	1	103	16.67	[60]	On-ramp before TLS
E_on_ramp_2	1	62	16.67	[60]	On-ramp after TLS
E_weaving	4	373	25.00	[90]	Weaving (bottleneck)
E_main_out	3	360	25.00	[90]	Downstream mainline
E_off_ramp	2	165	16.67	[60]	Exit ramp (Exit 11)

Table 3: Lane connections of the weaving conflict.

From edge	From lane	→	To edge	To lane
E_main_in	0	→	E_weaving	1
E_main_in	1	→	E_weaving	2
E_main_in	2	→	E_weaving	3
E_on_ramp_2	0	→	E_weaving	0
E_weaving	0	→	E_off_ramp	0
E_weaving	1	→	E_off_ramp	1
E_weaving	1	→	E_main_out	0
E_weaving	2	→	E_main_out	1
E_weaving	3	→	E_main_out	2

3.3.1 Krauss car-following model

The follower’s safe speed satisfies the Gipps safe-distance condition,

$$v_{\text{safe}} = v_{\text{lead}} + \frac{g - v_{\text{lead}} \tau}{\frac{v_{\text{follow}} + v_{\text{lead}}}{2b} + \tau}, \quad (4)$$

where g is the bumper-to-bumper gap, τ the desired headway/reaction time and b the comfortable deceleration. The realised speed is

$$v_{\text{next}} = \max[0, \min(v_{\text{des}}, v_{\text{safe}}, v_{\text{cur}} + a\Delta t) - \sigma \xi a \Delta t], \quad (5)$$

with stochastic imperfection $\sigma \in [0, 1]$, $\xi \sim U(0, 1)$, simulation step Δt , maximum acceleration a , and individual desired speed

$$v_{\text{des}} = v_{\text{limit}} \cdot \mathcal{N}(\mu_{\text{spd}}, \sigma_{\text{spd}}). \quad (6)$$

Calibration values per vehicle class are given in table 4.

Table 4: Krauss car-following parameters (speeds in m/s; km/h equivalents in brackets where useful).

Parameter	Car	Motorcycle	Bus	Truck
length (m)	4.5	2.2	12.0	8.0
minGap (m)	1.5	1.0	2.5	2.0
maxSpeed (m/s) [km/h]	27.0 [97]	25.0 [90]	25.0 [90]	28.0 [101]
a (m/s ²)	2.6	3.5	1.2	1.0
b (m/s ²)	4.5	5.0	4.0	3.5
σ	0.5	0.5	0.4	0.4
τ (s)	0.8	0.6	1.0	1.0
μ_{spd}	0.84	0.80	0.72	0.76
σ_{spd}	0.10	0.12	0.10	0.08

3.3.2 SL2015 sublane lane-changing model

Lane-change desirability is the sum of strategic, cooperative, speed-gain and keep-right utilities:

$$U = U_{\text{strategic}} + U_{\text{cooperative}} + U_{\text{speedGain}} + U_{\text{keepRight}}, \quad (7)$$

with

$$U_{\text{strategic}} \propto \frac{\text{lcStrategic}}{d_{\text{remaining}}}, \quad g_{\text{required}} = \frac{g_{\text{safe}}}{\text{lcAssertive}}. \quad (8)$$

We use $\text{lcStrategic}_{\text{car}} = 10$, $\text{lcAssertive}_{\text{car}} = 2$ (motorcycle 3.0, due to narrow width), and $\text{lcKeepRight} = 0$ for all classes (Singapore expressways do not enforce slow-lane return under congestion).

3.3.3 Demand and speed calibration

Demand was generated from per-minute YOLO counts; departure times were sampled uniformly within each 60 s interval. Table 5 lists the four routes and their vehicle counts.

Table 5: Routes and vehicle counts driving the SUMO simulation.

Route	Edges	Peak	Off-Peak
route_main_through	E_main_in → E_weaving → E_main_out	1,796	2,751
route_main_to_exit	E_main_in → E_weaving → E_off_ramp	1,343	1,475
route_ramp_merge	E_on_ramp → E_weaving → E_main_out	1,225	1,159
route_ramp_to_exit	E_on_ramp → E_weaving → E_off_ramp	129	111
Total	—	4,493	5,496

The desired-speed factor μ_{spd} was calibrated from off-peak Group 1 (fastest-lane) speed sampling in the YOLO data, which best approximates free-flow conditions. Identical values were used in both peak and off-peak runs so that performance differences come from demand alone.

3.3.4 Simulation configuration

$\Delta t = 0.5$ s, simulation duration 5,400 s, fixed seed 42. Per-edge metrics are written every 60 s through SUMO’s `edgeData` collector. The simulations were driven by the libsumo Python binding.

3.4 Peak Period Performance Parameters

The four parameters required by the brief are computed from the SUMO base-peak run (table 6). Figure 3 (a) plots the resulting density-flow scatter for each edge — the weaving edge sits clearly in the congested branch, while upstream and downstream mainline points cluster in the free-flow branch. Per-edge speed comparison (figure 4) confirms the weaving edge is the hot spot.

3.4.1 Average delay

$$T_{\text{ff}} = \frac{L_{\text{w}}}{v_{\text{ff}}} = \frac{373.01 \text{ m}}{25.0 \text{ m/s}} \approx 14.92 \text{ s.} \quad (9)$$

$$\bar{d} = \mathbb{E}[T_{\text{travel}}] - T_{\text{ff}} = 11.89 \text{ s/veh.} \quad (10)$$

3.4.2 Average queue length

$$Q = k \cdot L_{\text{w}} \cdot \left(1 - \frac{v}{v_{\text{ff}}}\right), \quad (11)$$

where k is density (veh/km) and the factor $(1 - v/v_{\text{ff}})$ isolates the congested fraction. Average $Q = 17.24$ veh; peak $Q = 29.77$ veh ($\approx 29.77 \times 5.6/100 \approx 167$ m).

3.4.3 Average time in queue

The average time spent at near-zero speed (below 0.1 m/s) on the weaving section was 0.46 s/veh.

3.4.4 Average excess accumulation (Little’s Law)

With arrival rate λ ,

$$N_{\text{opt}} = \lambda T_{\text{ff}}, \quad N_{\text{excess}} = N_{\text{actual}} - N_{\text{opt}}. \quad (12)$$

Average $N_{\text{excess}} = 17.24$ veh; peak 29.77 veh.

Table 6: Peak-period performance parameters on the weaving section (E_weaving), from base SUMO run.

Parameter	Average	Peak interval	Unit
Average delay	11.89	23.45	s/veh
Average queue length	17.24	29.77	vehicles
Time in queue	0.46	—	s/veh
Excess accumulation	17.24	29.77	vehicles
Net total time loss (network)	143396.6	—	veh-s
Avg weaving speed	48.96	—	km/h
Capacity (instantaneous)	6457.6	—	veh/h

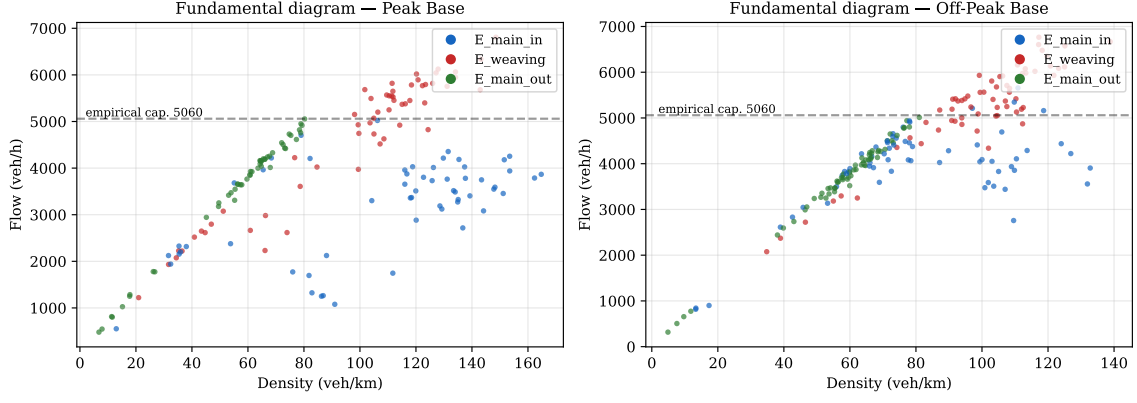


Figure 3: Empirical fundamental diagram per edge, peak vs off-peak. Red points ($E_weaving$) lie in the congested branch during peak; blue and green points stay in the free-flow branch. The dashed line marks the empirical capacity from equation (3).

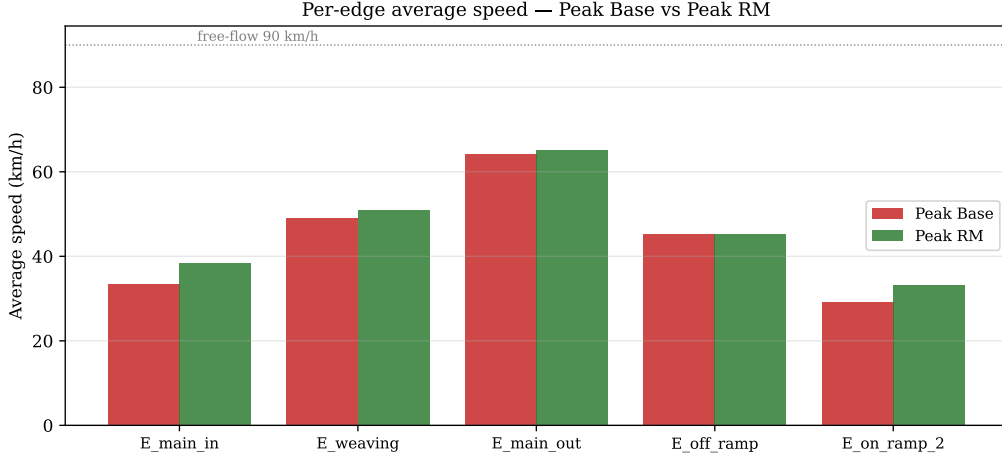


Figure 4: Per-edge average speed — the weaving edge is the slowest in both base and RM scenarios. RM brings the weaving speed up by about 4 km/h while leaving the upstream mainline largely unaffected.

4 Analysis During the Off-Peak Period

4.1 Conditions and capacity

The off-peak total upstream flow was 5,050 veh/h over 60 min, almost identical in magnitude to the peak. The differences are in composition: mainline flow 3,865 veh/h (up from 3,276 peak), exit flow 1,774 veh/h (down from 2,393), on-ramp flow 1,185 veh/h (down from 1,785); car exit ratio dropped from 51.2 % to 40.6 %. Off-peak car speeds averaged 61 km/h on the mainline, against 45 km/h during peak.

4.2 Off-peak performance parameters

The same four parameters are reported in table 7, drawn from the SUMO base-offpeak simulation.

Table 7: Off-peak performance parameters (E_weaving, base SUMO).

Parameter	Average	Peak interval	Unit
Average delay	9.31	15.43	s/veh
Average queue length	15.14	24.16	vehicles
Time in queue	0.078	—	s/veh
Excess accumulation	15.14	24.16	vehicles
Net total time loss (network)	89050.3	—	veh-s
Avg weaving speed	53.53	—	km/h
Capacity (instantaneous)	6752.6	—	veh/h

4.3 Peak vs off-peak — why the same flow gives different performance

Although the peak and off-peak total flows are within 0.2 % of each other, four mechanisms drive the peak-period degradation:

1. Exit demand share rises from 40.6 % to 51.2 %, more than half the cars must change leftward through 373 m.
2. On-ramp flow rises by 50 % (1,185 $\rightarrow$ 1,785 veh/h), all entering at lane 0 where exit-bound vehicles want to be.
3. Demand temporal concentration is higher (43 valid minutes vs 60), giving the weaving section less recovery time between platoons.
4. Composition shifts toward weaving traffic, fundamentally changing the section’s effective capacity. The simulation reproduces the speed drop 53.53 $\rightarrow$ 48.96 km/h with no change to infrastructure or vehicle parameters.

Figure 3 (b) shows the off-peak fundamental diagram: weaving points stay close to the free-flow branch except briefly during peak demand, while peak weaving points spread far into the congested branch.

5 Solutions to Improve the Operation

We tested three control strategies plus an upstream-VSL alternative, in both peak and off-peak demand profiles — 12 simulation runs in total. All runs share network, vehicle types, demand files and the fixed seed 42.

5.1 Variable Speed Limit (VSL)

Concept. VSL dynamically reduces the posted limit when downstream occupancy exceeds a threshold, so that flow remains below the capacity drop point and merges happen at lower speed differential. The control law is the simple thresholded version

$$v_{\text{lim}}(t) = \begin{cases} v_{\text{low}}, & \rho_w(t) > \theta_v \\ v_{\text{free}}, & \rho_w(t) < 0.7 \theta_v \end{cases} \quad (13)$$

with $v_{\text{free}} = 25$ m/s, $v_{\text{low}} = 12$ m/s, $\theta_v = 0.15$, evaluated every 30 s. Hysteresis ($0.7 \theta_v$) avoids chattering. Two placements were tested: *in-section* (E_weaving lane 0, the leftmost weaving lane that receives the merging stream) and *upstream* (E_main_in lanes 0–2, all three mainline lanes).

5.1.1 In-section VSL — result

In both periods the in-section VSL degrades performance (table 8). The mechanism is simple: lane 0 of E_weaving is the only lane connecting to E_off_ramp lane 0, so reducing its limit slows down exactly the queue of exit-bound vehicles, producing more spillback upstream. This is consistent with the literature finding that VSL in the bottleneck itself “*has limited or even no effect due to many mandatory lane-changing maneuvers near bottlenecks*” [5, 6].

Table 8: VSL (in-section, E_weaving lane 0) vs Base.

Metric	Peak			Off-Peak		
	Base	VSL	Δ	Base	VSL	Δ
Capacity (veh/h)	6457.6	6286.7	−2.6%	6752.6	6471.4	−4.2%
Avg delay (s/veh)	11.89	13.17	+10.8%	9.31	12.52	+34.5%
Queue (veh)	17.24	17.03	−1.2%	15.14	18.26	+20.6%
Time in queue (s/veh)	0.46	0.481	+4.6%	0.078	0.301	+286%
Excess acc. (veh)	17.24	17.03	−1.2%	15.14	18.26	+20.6%
Speed (km/h)	48.96	47.08	−3.8%	53.53	47.7	−10.9%
Net time loss (veh-s)	143396.6	173427.1	+20.9%	89050.3	167590.0	+88.2%

5.1.2 Upstream VSL — alternative placement

Following the literature recommendation, we relocated VSL onto E_main_in (all three lanes) so that arriving flow is harmonised *before* the weaving area. Same θ_v , same speed pair. The result (figure 5 and table 9) is much closer to neutral.

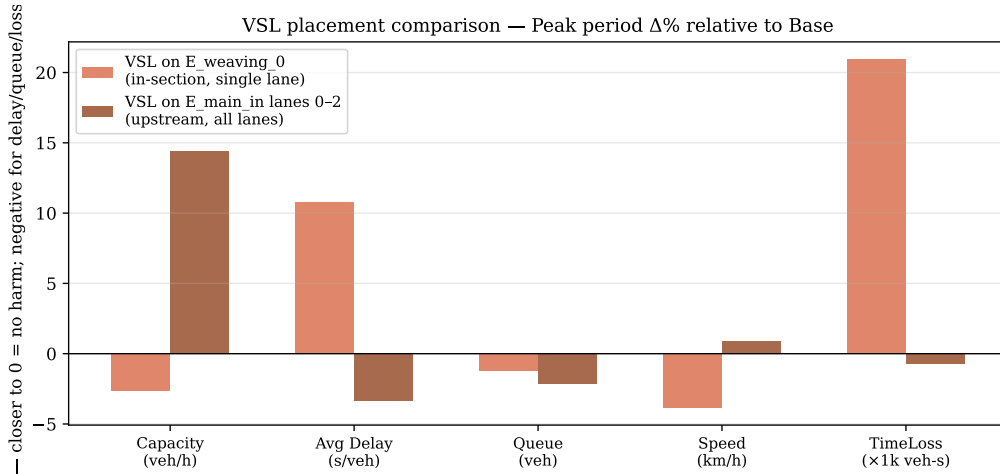


Figure 5: Δ vs Base for the two VSL placements during peak demand. Upstream placement (dark bars) avoids most of the harm caused by in-section placement (light bars).

Table 9: VSL placement comparison (E_weaving metrics).

Metric	Peak			Off-Peak		
	Base	VSL	VSL _{up}	Base	VSL	VSL _{up}
Capacity	6457.6	6286.7	7388.4	6752.6	6471.4	6862.2
Avg delay	11.89	13.17	11.49	9.31	12.52	9.78
Queue	17.24	17.03	16.87	15.14	18.26	15.8
Speed	48.96	47.08	49.41	53.53	47.7	52.49
Net time loss	143396.6	173427.1	142331.4	89050.3	167590.0	99602.3

The lesson is unambiguous: *placement matters* (in-section single-lane VSL is harmful, upstream all-lane VSL is approximately neutral) but *neither* placement matches the gain from ramp metering.

5.2 Ramp Metering (RM, ALINEA-style)

Concept. Ramp metering uses a traffic signal at the on-ramp to release vehicles in a controlled platoon, fragmenting the merging stream and protecting the mainline. We use a local feedback rule based on ALINEA [7, 8, 9]:

$$r(t + \Delta t) = r(t) + K_R [\hat{\rho}_{\text{des}} - \rho_w(t)], \quad (14)$$

where r is the metering rate, $\hat{\rho}_{\text{des}}$ the desired downstream occupancy, ρ_w the measured weaving occupancy, and K_R a feedback gain. In our microsimulation we approximate equation (14) by a discrete two-state controller: when $\rho_w(t) > \theta_r = 0.20$ the on-ramp signal switches from the default plan (81 G / 4 y / 5 r) to a metered plan (4 G / 2 y / 8 r) which admits roughly one vehicle per 14 s; the control is released when $\rho_w(t) < 0.7 \theta_r$. The threshold $\theta_r = 0.20$ falls within the 18–30% range typically used in European ALINEA field deployments.

5.2.1 RM result

Table 10 reports the per-period comparison. Peak network total time loss drops by 22.7%, average delay by 7.7%, and average weaving speed rises from 48.96 to 50.84 km/h. Off-peak gains are smaller because the section is already near free flow (53.53 km/h baseline).

Table 10: Ramp Metering vs Base.

Metric	Peak			Off-Peak		
	Base	RM	Δ	Base	RM	Δ
Capacity (veh/h)	6457.6	7209.3	+11.6%	6752.6	7402.0	+9.6%
Avg delay (s/veh)	11.89	10.97	−7.7%	9.31	8.65	−7.1%
Queue (veh)	17.24	15.7	−8.9%	15.14	13.54	−10.6%
Time in queue (s/veh)	0.46	0.22	−52%	0.078	0.053	−32%
Excess acc. (veh)	17.24	15.7	−8.9%	15.14	13.54	−10.6%
Speed (km/h)	48.96	50.84	+3.8%	53.53	55.12	+3.0%
Net time loss (veh-s)	143396.6	110833.5	−22.7%	89050.3	86585.3	−2.8%

The peak gain matches WSDOT field experience that turning ramp meters off increases travel time by about 22 % [9], providing external validation of the simulation result.

5.3 Combined VSL + Ramp Metering

Running both controllers simultaneously (table 11) does *not* improve over RM alone — the in-section VSL component drags the result down. Even with upstream-VSL+RM, the upstream speed reduction throttles the inflow that RM has already smoothed, giving a peak net time loss of 123997.1 veh-s vs 110833.5 for RM alone.

Table 11: Combined strategies vs RM alone.

Metric (Peak)	RM alone	VSL+RM	VSL _{up} +RM	Best
Avg delay (s/veh)	10.97	13.8	11.03	RM alone
Avg queue (veh)	15.7	17.23	15.46	VSL _{up} +RM
Speed (km/h)	50.84	45.77	50.75	RM alone
Net time loss (veh-s)	110833.5	173335.5	123997.1	RM alone

5.4 Strategy comparison and recommendation

Figures 6 to 10 consolidate the 12 scenarios. The improvement heatmap (figure 9) makes the conclusion immediate: only the RM column is uniformly green in the peak panel, while VSL columns swing into deep red on multiple metrics.

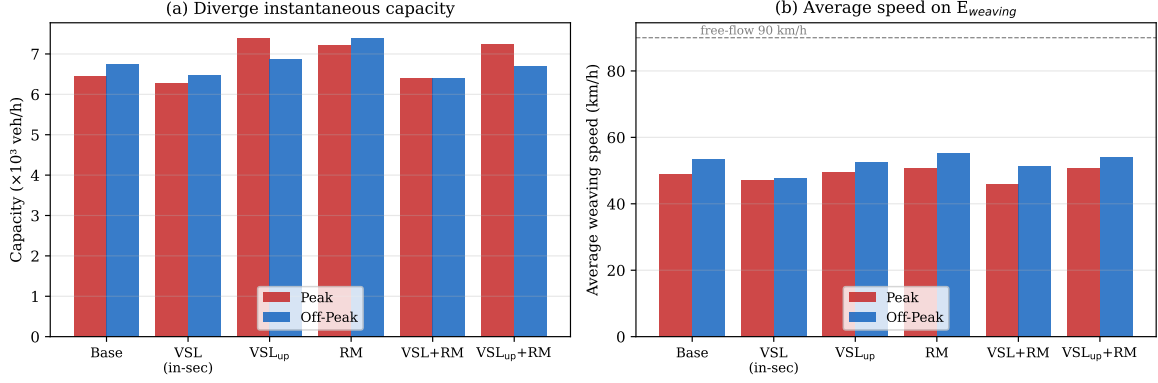


Figure 6: Diverge instantaneous capacity (a) and weaving section average speed (b) across 12 scenarios.

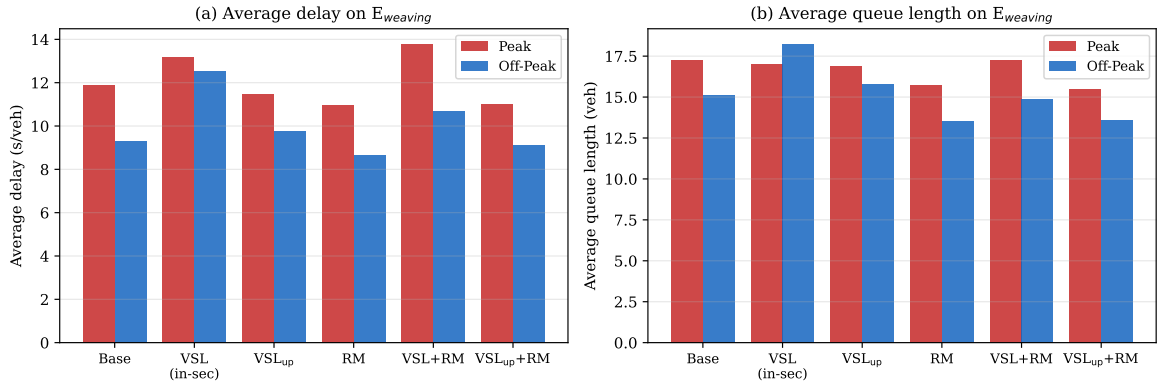


Figure 7: Average delay (a) and average queue length (b) on the weaving section across the 12 scenarios.

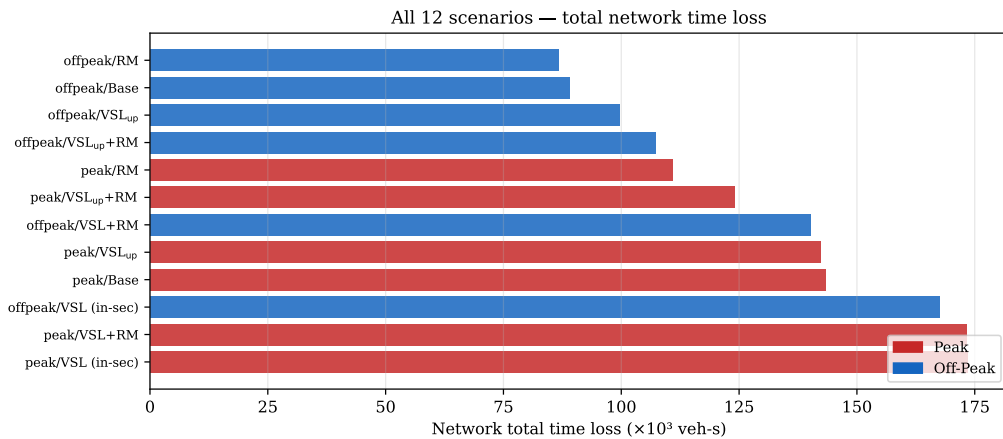


Figure 8: Network total time loss across all 12 scenarios. The two minima (RM peak, RM off-peak) are the only scenarios that improve over Base in both periods.

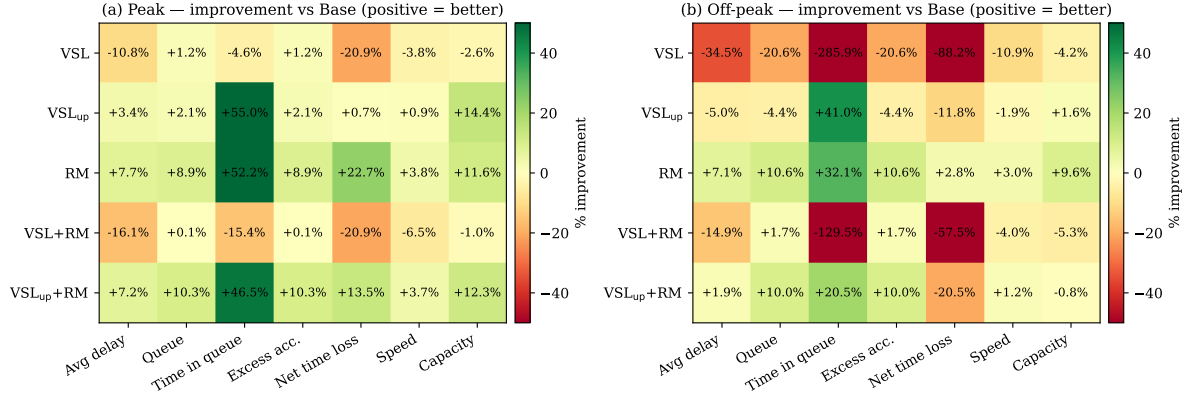


Figure 9: Strategy-vs-Base improvement heatmap (positive = better). Peak panel (left): RM is the only consistently green column; in-section VSL is consistently red. Off-peak panel (right): the picture is similar but smaller in magnitude as the system is already close to free flow.

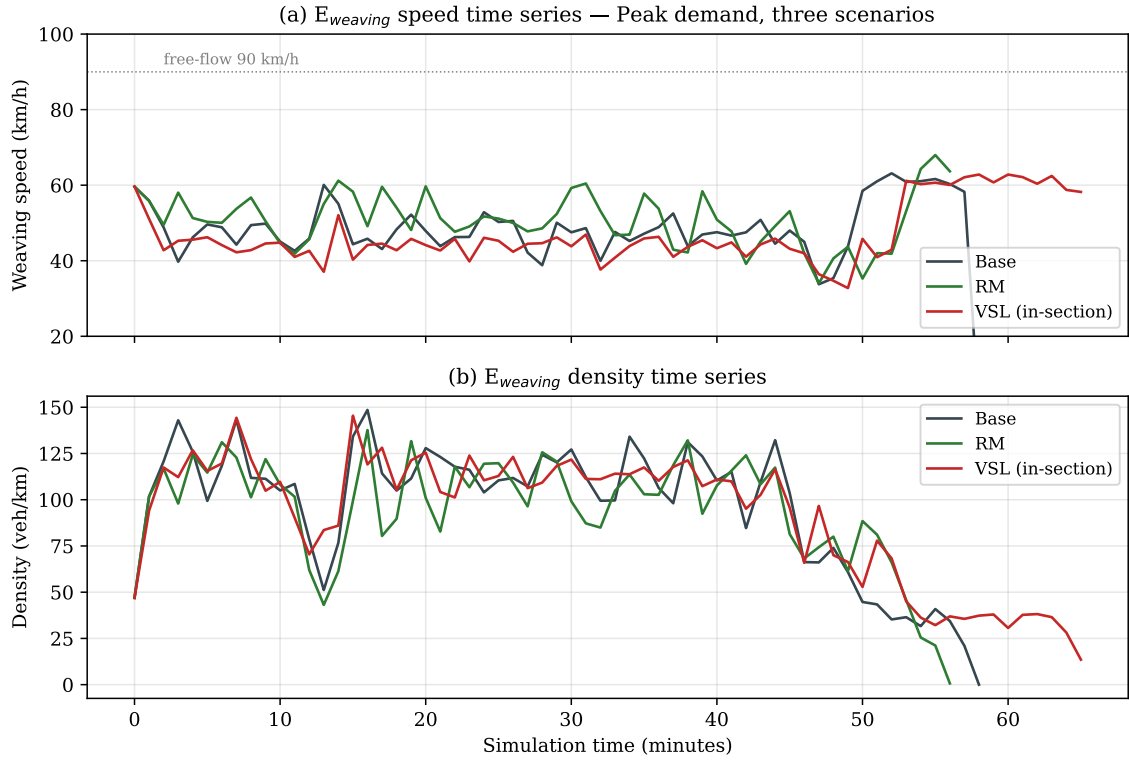


Figure 10: E_{weaving} time series under three peak scenarios. RM (green) suppresses the speed valleys around $t = 20$ min and reduces the density spikes between $t = 15$ and $t = 30$ min.

Table 12: Consolidated 12-scenario summary on the weaving section.

Metric	Peak						Off-peak		
	Base	VSL	VSL _{up}	RM	V+R	V _{up} +R	Base	VSL	VSL _{up}
Cap (veh/h)	6457.6	6286.7	7388.4	7209.3	6390.8	7253.4	6752.6	6471.4	6862.2
Delay (s/veh)	11.89	13.17	11.49	10.97	13.8	11.03	9.31	12.52	9.78
Queue (veh)	17.24	17.03	16.87	15.7	17.23	15.46	15.14	18.26	15.8
TimeQ (s/veh)	0.46	0.481	0.207	0.22	0.531	0.246	0.078	0.301	0.046
Excess (veh)	17.24	17.03	16.87	15.7	17.23	15.46	15.14	18.26	15.8
Speed (km/h)	48.96	47.08	49.41	50.84	45.77	50.75	53.53	47.7	52.49
TimeLoss (kveh-s)	143396.6	173427.1	142331.4	110833.5	173335.5	123997.1	89050.3	167590.0	99602.3

Recommendation. We recommend deploying *Ramp Metering alone* on the AYE westbound on-ramp upstream of the Exit 9/Exit 11 weaving section. RM provides a +11.6% instantaneous capacity gain, −22.7% network total time loss, and −7.7% average delay during peak; it is benign during off-peak. VSL — whether alone or combined with RM — is not recommended for this site, because the bottleneck is constrained by lane-change conflicts in the weaving section rather than by speed differential. If VSL is to be considered in future, it should be deployed upstream of E_main_in (not on a single in-section lane), combined with active lane-change guidance such as VMS-based lane recommendation [5, 6].

6 Conclusion

This project analysed a recurrent congestion bottleneck on the AYE westbound between Exit 9 and Exit 11. Field data — 43 minutes of peak and 60 minutes of off-peak traffic — were processed by a YOLOv11 vehicle-counting pipeline at four measurement points, yielding a peak total upstream flow of 5,060 veh/h and an off-peak flow of 5,050 veh/h. The empirical capacity of the 373 m weaving section is approximately 5,060 veh/h. Although total flows are nearly identical, peak traffic carries a much higher proportion of weaving traffic (51.2 % car exit ratio vs 40.6 % off-peak; on-ramp flow 1,785 vs 1,185 veh/h), generating the lane-change conflicts responsible for the observed speed drop and queueing.

A SUMO microsimulation calibrated to the YOLO speed and demand data was used to compute the four required performance parameters and to evaluate three control strategies (VSL, ramp metering, VSL+RM) plus an upstream-VSL alternative, in both peak and off-peak demand — 12 simulation runs in total (table 12). The peak base case shows average delay 11.89 s/veh, average queue 17.24 vehicles, average time in queue 0.46 s/veh, and average excess accumulation 17.24 vehicles on the weaving section, against 9.31 s/veh, 15.14 vehicles, 0.078 s/veh and 15.14 vehicles in off-peak.

Among the three strategies tested, only ramp metering improved performance consistently. Peak diverge capacity rose by +11.6%, average delay dropped by −7.7%, and total network time loss fell by −22.7%. In the off-peak period, where traffic is already near free flow, the improvement was modest. In-section VSL applied to the leftmost weaving lane increased delay and time loss in both periods because the bottleneck is constrained by lane-change conflicts, not speed; relocating VSL upstream removed most of the harm but did not match RM. These results align with empirical findings in the literature that lane-internal VSL on weaving sections has limited or negative throughput effect [5, 6], while local feedback ramp metering at 20% desired occupancy delivers travel-time savings on the order of 20% [7, 8, 9]. The recommended deployment is therefore Ramp Metering alone, with VSL kept as a future option only if combined with upstream placement and active lane-change guidance.

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